Name: Date: Period:

Newton #3 (Horizontal Net Force Problems)

Remember, all problems must include:

(1) A Picture (Free-body diagram) *(2) Knowns/Unknowns* *(3) Equation with letters*
(4) Plug & Chug *(5) Answer with Units*

$F_{NET} = ma$ $F_{NET} = \pm F_1 \pm F_2 \pm \cdots$ $F_G = mg$ $g = 9.8\,\text{m/s}^2$

1. You and a friend are able to use a rope to pull a giant cement statue across the ground with an acceleration of 0.07 m/s^2. If you apply a force of 500 N, your friend applies a force of 480 N, and the force of friction on the block is 300 N, what is the mass of the statue?

2. Neal is stealing a refrigerator (173 kg) from the bank. If the force of friction on the refrigerator is 254 N, what force must he apply to have an acceleration of 1.27 m/s^2?

3. A loaded semi truck has a mass of 36 000 kg. Its engine applies a forward force of 20 820 N, causing it to accelerate at $0.12\,\text{m/s}^2$. Calculate the force of friction on the truck.